

FORT MADISON, IA, USA

WMO: 725483

Lat: 40.659N

Lon: 91.327W

Elev: 724

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.2	(c) 4.9	(d) -8.4	(e) 3.6	(f) 1.6	(g) -4.1	(h) 4.5	(i) 6.6	(j) 26.9	(k) 24.0	(l) 24.4	(m) 23.2	(n) 7.3	(o) 310	(p) 0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	93.0	78.1	90.6	77.3	88.3	75.7	81.4	89.7	79.3	87.5	77.7	85.9	7.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	155.6	86.7	76.9	144.1	85.2	74.9	134.4	83.1	45.6	90.5	43.5	87.8	41.5	86.6	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	19.0	17.0	DB	-7.5	96.4	6.6	3.3	-12.2	98.8	-16.1	100.7	-19.8	102.5	-24.5	104.9
			WB	-6.6	83.4	7.5	1.2	-12.0	84.3	-16.4	85.0	-20.7	85.6	-26.1	86.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	26.0	29.2	41.9	53.2	64.4	72.9	76.4	74.8	68.0	55.8	42.0	31.9
	DBStd	20.04	12.15	11.41	11.74	10.16	8.57	6.14	5.82	5.55	7.90	9.64	10.09	10.55
	HDD50	2614	745	585	308	79	4	0	0	0	0	51	277	565
	HDD65	5522	1209	1002	721	371	119	10	1	1	57	314	691	1026
	CDD50	3773	1	3	56	176	451	687	817	769	541	231	37	4
	CDD65	1206	0	0	4	18	102	246	354	305	147	29	1	0
	CDH74	11164	0	0	26	172	862	2208	3581	2755	1333	224	3	0
	CDH80	4110	0	0	1	31	256	799	1452	1049	468	54	0	0
(14) Wind	WSAvg	7.5	8.9	8.9	9.3	9.5	7.8	6.5	5.1	4.5	5.4	7.0	8.5	8.4
(15) Precipitation	PrecAvg	37.5	1.8	1.7	2.6	3.8	4.8	4.6	4.1	3.9	3.0	2.8	2.5	1.8
	PrecMax	51.6	3.2	4.9	5.4	6.1	11.5	10.3	11.7	7.5	6.0	7.3	6.9	4.4
	PrecMin	27.6	0.5	0.2	0.7	2.2	0.8	0.7	1.5	1.2	0.7	1.1	0.0	0.4
	PrecStd	7.0	0.8	1.2	1.5	1.1	2.9	2.6	2.7	2.0	1.8	1.7	1.7	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	62.8	78.7	84.4	90.6	93.1	97.3	97.0	93.2	86.2	72.5	63.0
		MCWB	52.0	52.3	63.8	67.4	74.0	77.0	79.1	76.7	75.1	71.4	58.0	56.8
	2%	DB	52.2	54.9	71.7	79.3	86.0	90.5	93.2	92.5	89.8	80.6	66.0	55.0
		MCWB	47.0	47.5	59.9	62.9	71.9	76.4	79.4	77.8	74.2	65.9	56.9	49.9
	5%	DB	46.0	49.6	64.4	74.6	82.0	88.1	90.6	89.9	85.8	75.0	62.5	51.0
		MCWB	41.4	44.3	56.2	60.7	68.2	74.8	78.8	77.5	73.1	63.7	54.1	47.2
	10%	DB	41.4	45.0	59.4	70.3	79.1	84.4	88.1	86.0	81.7	71.7	56.9	45.8
		MCWB	36.9	40.5	51.6	58.5	66.0	73.6	77.0	75.3	69.8	62.2	49.7	41.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	53.8	65.3	69.4	77.1	81.7	84.5	83.2	79.2	73.3	61.8	58.8
		MCDB	57.2	60.3	75.3	80.9	87.2	89.6	92.2	92.6	87.3	83.0	66.5	62.9
	2%	WB	46.5	48.8	61.5	65.6	73.6	78.8	82.5	80.2	76.8	69.7	58.0	51.8
		MCDB	50.4	54.8	68.0	75.4	82.9	86.9	90.2	89.3	85.6	75.3	64.3	54.1
	5%	WB	41.8	44.4	58.0	63.1	71.2	77.0	80.0	78.3	74.6	66.2	54.2	46.4
		MCDB	47.1	48.6	64.3	71.2	79.9	85.1	87.9	86.7	82.3	73.0	60.6	49.3
	10%	WB	37.7	40.2	52.8	60.1	68.6	75.0	78.0	76.4	72.4	62.7	50.3	42.2
		MCDB	41.1	44.7	58.9	68.9	76.0	82.6	86.0	83.9	79.6	69.7	56.4	46.0
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.1	17.5	19.5	19.1	18.2	18.4	18.9	21.4	19.1	16.2	13.7
		MCDBR	20.0	21.5	23.3	25.5	22.4	20.7	19.3	21.7	22.7	23.3	21.7	18.9
	5% WB	MCWBR	16.0	16.0	14.9	13.7	11.4	10.6	10.5	12.1	10.9	12.7	15.0	15.6
		MCWBR	19.1	19.6	21.4	22.1	19.9	18.4	18.1	18.9	19.5	18.8	19.9	18.3
(40) Clear-Sky Solar Irradiance	taub		0.290	0.308	0.339	0.389	0.422	0.425	0.443	0.433	0.395	0.349	0.302	0.289
		taud	2.453	2.400	2.404	2.266	2.236	2.284	2.207	2.255	2.312	2.443	2.515	2.473
	Ebn at Noon		278	288	290	280	272	271	264	264	266	269	272	269
		Edh at Noon	24	30	34	42	44	42	45	42	36	29	23	22
(44) All-Sky Solar Radiation	RadAvg		585	886	1176	1512	1767	1978	1976	1737	1492	997	670	498
	RadStd		69	75	83	120	126	185	126	95	109	110	54	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page